

Integration mechanisms and R&D project performance[☆]

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Abstract

Information processing theory suggests the need for different types of integration mechanisms in R&D project management depending on levels of uncertainty and equivocality. This paper examines the use of these mechanisms and their links to project performance in a sample of 121 R&D projects in a large research laboratory. Overall, it is found that formal leadership, planning and process specification, and to a lesser extent information technology use are related to project performance while the positive effects of horizontal structures are apparently balanced out by their costs. The integration mechanisms studied act on performance partly through their effect on horizontal communications. Modest support was found for the contingency hypotheses derived from information processing theory. It appears that managers adjusted their use of horizontal structures, planning and process specification, and informal leadership to project uncertainty but not to project equivocality. The positive effects of horizontal communications on performance were found to be greatest under high project equivocality as would be predicted by information processing arguments. Moreover, with the exception of formal leadership, the use of integration mechanisms did not enhance performance in contexts of low uncertainty and low equivocality. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Information processing theory; R&D project management; Integration mechanisms; Innovation management

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1. Introduction

In an innovative process, distinct sets of specialized skills rarely found within one individual are often necessary to accomplish the complex tasks required for the development, manufacturing and marketing of a product (Allen, 1986; Roberts, 1988; Von Hippel, 1990). At the same time, communication, collaboration and integration between specialists and functions are required to maximize the synergy between the various interdependent parts of a project (Brownlie, 1987; Moenaert and Souder, 1990; Ettlie and Reza, 1992; Hitt et al., 1993). As Dougherty (1992) shows, integration tends to be hampered by the development of distinct “thought worlds” associated with individual specialization. This tension between the need for differentiation and for integration lies at the heart of the study of organizations and seems particularly salient to the problem of innovation (Thompson, 1967; Lawrence and Lorsch, 1969; Daft and Lengel, 1986).

To solve this problem, a number of different approaches to integration and coordination among the diverse specialists involved in R&D project development have been proposed. These mechanisms can include formal or informal project leadership, formal planning procedures, horizontal structural or cultural mechanisms, and the use of communication technologies (Moenaert and Souder, 1990; Hitt et al., 1993; Adler, 1995).

Integration mechanisms are not costless however, (Hitt et al., 1993; Emmanuelides, 1993). For example, regular project meetings and status reports absorb time and energy that might be better spent in execution. Moreover, the effectiveness of the various mechanisms is likely to vary depending on specific project needs. In particular, larger and more complex projects seem likely to warrant more elaborate mechanisms. Similarly, radical innovations may require different types of organizational arrangements from more incremental projects. In this paper, we use information processing theory (Daft and Lengel, 1986) to derive hypotheses concerning the prevalence and efficacy of five types of integration mechanisms under different project conditions. These are tested on a sample of projects from a large R&D laboratory (labeled Centerlab).

2. Theoretical base: information processing requirements and integration mechanisms

2.1. *Uncertainty, equivocality and information processing needs*

Daft and Lengel (1986) propose that two distinct task dimensions determine organizational information processing needs in any given situation: uncertainty and equivocality. Uncertainty refers to the absence of answers to well-defined questions: for example, a lack of knowledge about future events or the consequences of specific actions. Equivocality refers to a deeper level of ambiguity and confusion concerning the nature of the questions asked, the goals pursued and the problems to be solved (Daft and Lengel, 1986). While the collection of hard factual data can help resolve uncertainty, it may be of relatively little help under equivocality when the nature of the task itself is unclear.

Rather, the participants in the situation will need to exchange ideas and interpretations to develop a common understanding of the problem. Equivocality and uncertainty often occur together. However, it is also conceivable that a project may involve a relatively small number of simple and familiar components (i.e., low uncertainty) and yet be difficult to get off the ground because of ambiguities concerning what is to be accomplished (high equivocality). Similarly, a project may be well-defined (low equivocality) but require the collection of a large amount of information from a variety of sources to proceed (high uncertainty).

R&D projects clearly involve variable levels of uncertainty and equivocality depending on their size, complexity and novelty to the organization (Schrader et al., 1993). More specifically, as explained in the next paragraphs, several previous researchers on innovation have noted that Perrow's (1967) technological dimensions of task variety and task analyzability can be considered to reflect the uncertainty and equivocality dimensions of R&D projects (Tushman and Nadler, 1978; Daft and Lengel, 1986; Gales et al., 1992; Keller, 1994).

Task variety is defined as the number of unrelated tasks or unexpected events associated with the accomplishment of a project (Withey et al., 1983; Daft and Lengel, 1986; Gales et al., 1992). Greater task variety creates greater uncertainty because there are more sources of unpredictability and more potential disruptions associated with the coordination of interdependent pieces of the project. Thus, there is an ongoing need to collect and share information to work out how to execute the next task, find out what is going on elsewhere, deal with disruptions and ensure that projects are on track (Daft and Macintosh, 1981; Keller, 1994). In contrast, when task variety is low, work can be pre-programmed, people know what others are doing and there is less need to collect information (Daft and Macintosh, 1981). Task variety is related to what Emmanuelides (1993) called "internal uncertainty" stemming from the non-routine nature of R&D tasks and the high levels of technical and organizational interdependence involved in their execution. As Gales et al. (1992) note, dimensions related to uncertainty and task variety have been proposed in many innovation studies although operationalizations vary (e.g., Withey et al., 1983). In this research, task variety at the project level is operationalized as a function of the number of people involved and the degree of administrative and technological complexity (see Section 4).

Task analyzability is defined as the degree to which a computational process can be used to develop solutions to problems (Daft and Lengel, 1986; Gales et al., 1992), and is usually associated with familiarity and the use of established procedures and practices (Withey et al., 1983). Incremental projects that involve known techniques and applications will be relatively analyzable as it will be possible to decompose them into a series of stages or activities based on past experience. Conversely, more radical projects involving unfamiliar technology and unclear application requirements will be low in analyzability because of the lack of reference points for defining the steps to be followed (Fischer, 1979; Gales et al., 1992). Therefore, there will be ambiguity surrounding the nature of the tasks to be accomplished and the problems to be solved. Thus, low analyzability is closely associated with high equivocality (Daft and Lengel, 1986). Following other researchers (Tushman and Nadler, 1978; Gales et al., 1992; Green et al., 1995), we operationalize analyzability in terms of the inverse of the degree of radicality

of the innovation as well as in terms of knowledge about how to approach the design process (see Section 4). Again, this dimension has appeared in various guises in previous research on innovation as a way to take into account the exceptionally ambiguous nature (equivocality) of some R&D projects (e.g., Ettlie et al., 1984; Dewar and Dutton, 1986; Gales et al., 1992), an ambiguity that goes beyond a simple lack of data to include a lack of knowledge about cause–effect relationships and objectives (Thompson, 1967; Daft and Lengel, 1986).

2.2. Information processing requirements and integration mechanisms

As we noted in the introduction, successful innovation requires the collaboration of many different specialists whose work must be integrated to create a coherent whole. A variety of different integration mechanisms can be used to achieve this. From an information processing perspective, these mechanisms can be seen as means for reducing either uncertainty or equivocality. Daft and Lengel (1986) examine the types of communication media and structural mechanisms that favor each of these objectives. They argue that while formal rules and regulations, information systems and special reporting procedures are capable of handling large quantities of factual information and, thus, of reducing high levels of uncertainty, these mechanisms will be inadequate under conditions of high equivocality. In such situations, “rich” information media such as face-to-face meetings and personal contact are required to provide opportunities for people to share understandings, debate issues, and develop consensus on problem definitions (Daft and Lengel, 1984, 1986). Specifically, Daft and Lengel (1986) identify group meetings and integrator roles as structural mechanisms that can permit the reduction of equivocality through close interpersonal contact.² They see formal planning systems as intermediate mechanisms that may assist in both the reduction of uncertainty and equivocality. For example, they argue that planning begins with face-to-face meetings to establish objectives and a general course of action, thus reducing equivocality. Subsequently, plans become a data processing and control device more closely associated with uncertainty reduction.

Therefore, Daft and Lengel’s (1986) theory suggests that the most appropriate integration mechanisms will vary depending on levels of project uncertainty and equivocality. This research aims to test this proposition. Although some previous researchers have examined the relationship between uncertainty and equivocality of R&D projects and the amounts and richness of information exchanged (e.g., Keller, 1994; Gales and Mansour-Cole, 1995; Gales et al., 1992), to our knowledge, this is the first study to relate the theory directly to mechanisms for integration. In this study, we explicitly consider five possible types of integration mechanisms: horizontal structures, formal leadership (project management); planning and process specification; use of information technologies, and informal leadership (project champions). We also look at the role of one intermediate indicator of integration effort: horizontal communication.

² Note that face-to-face contact can lead to conflict as well as consensus. Nevertheless, it can be argued that the surfacing of conflict is a necessary first step to the reduction in equivocality. Certain strategic decision making techniques (e.g., dialectics) are based on this principle (e.g., Mitroff et al., 1979).

A set of three hypotheses are formulated for each mechanism. The first hypothesis in each set (H1a–H6a) concerns the overall relationship between the use of the mechanism and performance. The other two hypotheses are contingency hypotheses concerning the appropriateness of the mechanism in different contexts. Four types of project contexts are considered (see Table 1), corresponding to a 2×2 matrix in which the dimensions represent uncertainty (high or low) and equivocality (high or low). This formulation corresponds to that of Daft and Lengel (1986).

The two contingency hypotheses formulated for each integration mechanism reflect two different forms of contingency theory. Van de Ven and Drazin (1985) note that contingency theories can be founded on several different conceptual approaches. One approach called the “selection” approach (Van de Ven and Drazin, 1985) is based on the idea that organizations will tend to modulate their behavior depending on the specific context. This may occur because managers systematically choose the most efficient approaches in each situation or because natural selection eliminates organizations that do not make the most efficient choices. In either case, this approach implies that behavior–context combinations which are most effective will be more frequent than ineffective combinations. Our second series of hypotheses for each integration mechanism (H1b–H6b) reflect this idea. Specifically, when information processing theory suggests that a mechanism is more appropriate in some contexts than others (defined in terms of uncertainty and equivocality), we expect the intensity of use of the mechanism to be higher in those cells.

An alternative approach to contingency theory is called the “interaction” perspective by Van de Ven and Drazin (1985). This approach posits that the closer the correspondence between context and behavior, the higher the performance. In this version of the theory, managers imperfectly adjust their behavior to the project context either because they are unaware of contingent requirements, or because they take into account concerns other than efficiency. The result is that performance varies as a function of the degree of fit. In our case, this approach is reflected in the third hypothesis presented for each mechanism (H1c–H6c). Specifically, when information processing theory indicates that a mechanism is more appropriate in a given context, the degree of association between the use of this mechanism and project performance is also expected to be higher. We now present our specific arguments for each of the selected integration mechanisms, drawing on information processing theory (Daft and Lengel, 1986) and on related innovation literature.

Table 1
Uncertainty, equivocality and hypothesized integration mechanisms

	Low uncertainty (low task variety)	High uncertainty (high task variety)
High equivocality (low task analyzability)	horizontal structures , formal leadership, planning, informal leadership , horizontal communication	horizontal structures , formal leadership , planning , information technology , horizontal communication
Low equivocality (high task analyzability)	informal leadership	formal leadership, planning, information technology , horizontal communication

3. Research hypotheses

3.1. Horizontal structures

Horizontal structures overlay traditional “functional” structures. They lie on a continuum ranging from informal mechanisms such as temporary assignments and task forces in which functional authority continues to be dominant to matrix organizations in which dual lines of authority are permanently embedded to organizations in which cross-functional project teams are the dominant unit and functions play a staff role (Ford and Randolph, 1992). These mechanisms have long been associated with effective innovation in the organization theory literature (Galbraith, 1973; Mintzberg, 1979). They are intended to counteract the centrifugal forces created by high levels of functional specialization among participants in innovation projects by forcing face to face contact, mutual adjustment and lateral communication (e.g., Hitt et al., 1993; Keller, 1994; Adler, 1995). By improving the amount and quality of information exchange at the project level, these mechanisms also enable resources to be quickly reallocated as needs change over time (Larson and Gobeli, 1988). The use of cross-functional teams has been shown in prior research to be strongly related to project performance (Zirger and Maidique, 1990; Brown and Eisenhardt, 1995). Therefore, as a working hypothesis, we predict that the relationship between the use of horizontal structures and performance will be positive.

These structures are not without disadvantages, however (Ford and Randolph, 1992; Hitt et al., 1993). While intended to improve communications and coordination, they can be costly and time-consuming (Larson and Gobeli, 1988). Dual lines of authority can lead to new forms of stress (Nonaka, 1990) including ambiguity about responsibilities, and conflict between functional and project managers on issues such as project priorities, schedules and resources (Barker et al., 1988; Ford and Randolph, 1992; De Laat, 1994). Thus, these structures may not always be beneficial. Potential communication benefits must be traded off against the costs involved.

According to Daft and Lengel (1986), the communication benefits of mechanisms that favor face to face contact (such as horizontal structures) are greatest under high equivocality. Face-to-face contact allows immediate feed-back and adjustment and can incorporate a wide variety of informational cues (including content, verbal tone, body language, etc.). This is an effective approach when people need to work together to clarify what they are doing (Daft and Lengel, 1984) but may be less necessary when project parameters are familiar and clear (i.e., under low equivocality). Adler (1995) also argues that interactive mechanisms such as teams are optimal for highly novel projects but too costly for more routine ones. Therefore, we expect that horizontal structures will be more frequently used and more effective in the case of projects of high equivocality and they will be less prevalent and less effective in contexts of low equivocality. The following three hypotheses summarize the arguments of the last three paragraphs.

Hypothesis 1a. *The use of horizontal structures will be positively related to project performance.*

Hypothesis 1b. *Horizontal structures will be used to a greater extent for projects of high equivocality than for those of low equivocality.*

Hypothesis 1c. *The use of horizontal structures will be more strongly related to project performance under high equivocality than under low equivocality.*

3.2. Formal leadership

Formal project leadership plays a key integrative role in innovation projects both within the project team and between the team and outsiders. As Brown and Eisenhardt (1995) indicate in their extensive review of the product development research literature, powerful project leadership is strongly associated with effectiveness. In their internal integrator role, project leaders can help motivate team members to work together to resolve differences and work out problems, relying on their authority, expertise and personal credibility. In their external integrator roles, formal project leaders act as a “bridge between the project team and senior management” (Brown and Eisenhardt, 1995). The greater the power accorded to formal project managers, the greater will be their capacity to mobilize resources and create a cohesive team (Ancona and Caldwell, 1990, 1992; Clark and Fujimoto, 1991). In addition as Emmanuelides (1993: 378) suggests and as Ancona and Caldwell (1992) confirm in their research, “Effectiveness of a product development team often hinges on its ability to buffer certain design activities against outside interference.” Leaders with authority over all aspects of the project will be better able to manage the interface with top management as well as internal team dynamics.

Formal project leaders, as “integrators” in Daft and Lengel’s (1986) terminology, contribute to both uncertainty and equivocality reduction. They reduce uncertainty by acting as a formally recognized conduit for data transmission between team members and between the team and top management. However, their most important role is likely to be in equivocality reduction. They are in a privileged position to interpret top management concerns and to organize and arbitrate face-to-face discussions and negotiations among team members concerning the project. Therefore, the reliance on formal project leadership is likely to be more prevalent and more crucial for performance for projects involving both high uncertainty and high equivocality. The following hypotheses summarize our arguments here.

Hypothesis 2a. *The use of formal project leadership will be positively related to project performance.*

Hypothesis 2b. *The use of formal project leadership will be highest for projects of high equivocality and high uncertainty and lowest for projects of low equivocality and low uncertainty.*

Hypothesis 2c. *The association between the use of formal project leadership and performance will be strongest for projects of high equivocality and high uncertainty and weakest for projects of low equivocality and low uncertainty.*

3.3. Planning and process specification:

Many writers have noted that careful planning and process specification can improve project performance. Indeed, several project management techniques have been designed based on the idea that by breaking down projects into their component parts and projecting timelines and milestones for each element, it is possible to better coordinate the whole (Moenaert and Souder, 1990; Von Hippel, 1990; Wheelwright and Clark, 1992; Cooper, 1996). It can be argued that when everyone knows exactly what they have to accomplish and when, the need for ongoing mutual adjustment is lessened, improving communication and performance. Thus, planning and process specification is a potentially powerful integration mechanism and several empirical studies have confirmed its role in improving project performance (Cooper and Kleinschmidt, 1987; Zirger and Maidique, 1990). On the other hand, another current of literature is less enthusiastic about the benefits of planning. For example, Mintzberg (1994) suggests that the creative process of innovation and the constraints of formal planning are to some extent incompatible and Eisenhardt and Tabrizi (1995) found that planning tended to slow down product development. This might lead us to question the role of formal planning, especially under conditions of equivocality where radically new innovations are being considered. Based on data concerning the design–manufacturing interface in 13 firms, Adler (1995) argued that planning mechanisms are optimal at moderate levels of novelty but less appropriate for extremely novel or for routine projects.

Daft and Lengel (1986) suggest that “planning is a dynamic process that includes elements of both equivocality reduction and data sharing.” As noted above, they view the initial stages of the planning process as a means to reach consensus through the exchange of rich information, with the more bureaucratic and uncertainty reducing elements dominating later in the process. Based on this logic and encouraged by the weight of empirical evidence, we adopt the working hypothesis suggested by Daft and Lengel: that planning and process specification is generally beneficial, but most useful under conditions of high uncertainty and equivocality. An alternative hypothesis consistent with Mintzberg’s (1994) position would associate it mainly with uncertainty, while Adler’s (1995) proposal would suggest a curvilinear relationship with equivocality. We return to these issues in the conclusions.

Hypothesis 3a. *The use of formal planning and process specification will be positively related to project performance.*

Hypothesis 3b. *The use of formal planning and process specification will be highest for projects of high equivocality and high uncertainty and lowest for projects of low equivocality and low uncertainty.*

Hypothesis 3c. *The association between the use of formal planning and process specification and performance will be strongest for projects of high equivocality and high uncertainty and weakest for projects of low equivocality and low uncertainty.*

3.4. Information technology

The use of information technology is expected to facilitate coordination during innovation projects by increasing the speed of communication, reducing its cost, widening the scope of the information network and allowing access to communal information (Dean and Snell, 1991; Hitt et al., 1993; Adler, 1995; Fulk and DeSanctis, 1995). Useful technologies in this respect may include electronic mail, voice mail, fax, decision support systems, CAD/CAM, and electronic data management. For example, CAD/CAM technologies allow easy shared access to ongoing product designs (Adler, 1995), while electronic communications facilitate horizontal communications across intra-organizational and inter-organizational boundaries, lessening the need for physical proximity (Fulk and DeSanctis, 1995).

Daft and Lengel (1986) view information technology in general as a medium highly appropriate for uncertainty reduction, but less useful for rich communication and, therefore, not strongly associated with equivocality reduction. This is the form of the hypothesis that we have adopted here. It should be noted, however, that there is considerable controversy in the literature over the role of newer technologies such as electronic mail and voice mail. For example, while some authors have confirmed the low rating of electronic mail for equivocality reduction (Schmitz and Fulk, 1991), others have suggested that it can be used to communicate richer information (Kiesler, 1986; Rice and Gail, 1989). Markus (1994) argues that newer communication media are not inherently “rich” or “lean” but that different social groups can construct them so that they are used in richer or more structured ways. For example, she showed that while electronic mail does not, as such, appear to be open to the flexible formulation of thoughts and immediate feedback that can be gained in face-to-face conversation, certain organizational groups may come to adopt them for use in this way. Thus, the potential for equivocality reduction via the use of technology may be socially constructed and context-dependent (see also Zack and McKenny, 1995). Even so, these asynchronous technologies do not seem to be inherently favorable to rich exchanges to the same extent as telephone calls or personal and group meetings (Hinds and Kiesler, 1995). Therefore, we view them here mainly as tools for uncertainty reduction.

Hypothesis 4a. *The use of information technologies will be positively related to project performance.*

Hypothesis 4b. *Information technologies will be used to a greater extent for projects of high uncertainty than for those of low uncertainty.*

Hypothesis 4c. *The use of information technologies will be more strongly related to project performance under high uncertainty than under low uncertainty.*

3.5. Informal leadership

This refers to the use of project champions who may not necessarily hold formal leadership positions. Technical champions who take to heart the success of the project

and devote energy to convincing others have been found to contribute to project performance in a number of empirical studies (Cooper, 1990; Hise et al., 1990; May and Pearson, 1993). Howell and Higgins (1990) show that champions demonstrate many of the characteristics and behaviors associated with transformational leadership (risk-taking, persistence, persuasiveness, influence attempts), yet do this informally. Therefore, we argue that champions act as informal “integrators”. As “integrators”, their role can be seen mainly as one of equivocality reduction through direct contact with others (Daft and Lengel, 1986; Lee and Na, 1994). Through their persuasive powers and their willingness to commit to the project, they assist in creating convergence and enthusiasm around common goals. In addition, their enthusiasm may stimulate support for the project from outside the team. However, it can be argued that project champions’ scope for influence is usually more limited than that of formal project managers, especially when task variety (and thus uncertainty) is high. This is because of the difficulty of gaining acceptance among a large number of project participants with diverse expertise without formal authority and responsibility for the project. Therefore, we predict that informal leadership will be most prevalent and effective under conditions of high equivocality and low uncertainty.

Hypothesis 5a. *The presence of project champions will be positively related to project performance.*

Hypothesis 5b. *The presence of project champions will be highest for projects of high equivocality and low uncertainty and lowest for projects of low equivocality and high uncertainty.*

Hypothesis 5c. *The association between the presence of project champions and performance will be strongest for projects of high equivocality and low uncertainty and weakest for projects of low equivocality and high uncertainty.*

3.6. Horizontal communication

The importance of effective horizontal communication for innovation success is implicit in most of the discussion so far. It has also been firmly established in a major stream of previous research (Tushman, 1979; Allen, 1986; Gales et al., 1992; Brown and Eisenhardt, 1995; Moenaert et al., 1995). Zirger and Hartley (1994) also place information processing capacity as a mediating variable at the center of their model of project development speed. Following previous work, we argue that the effectiveness of horizontal communications is an intermediate variable that reflects the extent to which satisfactory integration has been achieved between the various organization members associated with a project.

Although effective communication may be more difficult to achieve for more uncertain projects because of the greater number of people and types of expertise involved, it is likely that it will be particularly critical for such projects, given the greater need for integration of diverse viewpoints. Similarly, we expect that high

equivocality will create a need for intensive and effective horizontal communication between project team members. Indeed, Morelli et al. (1995) show that communication needs for coordination can be predicted based on the characteristics of the project using “design structure matrices” that represent the complexity of the task. They argue further that this type of analysis should be used to identify appropriate integration mechanisms for project success. Keller (1994) showed that a fit between the amount of information processing in R&D project groups and task variety had a positive effect on performance, although the fit between information amount and unanalyzability (equivocality) was unrelated to performance in their sample. Gales et al. (1992) provide further empirical evidence that managers adjust the amount and richness of communication depending on uncertainty and equivocality and indicate that this adjustment is positively related to performance.

Because horizontal communication is seen as a mediating variable between integration mechanisms and performance, and because our measures reflect communication effectiveness rather than amounts, our three hypotheses for this mechanism take a somewhat different form from that used for the five integrating mechanisms. We first hypothesize a direct association between effective communication and performance. Then we hypothesize a mediating effect of horizontal communications between integration mechanisms and performance. Finally, we predict that uncertainty and equivocality will moderate the project performance consequences of effective horizontal communications, as argued in the preceding paragraph.

Hypothesis 6a. *Effective horizontal communication will be associated with higher project performance.*

Hypothesis 6b. *Horizontal communications effectiveness will act as a mediating variable between integration mechanisms and project performance.*

Hypothesis 6c. *The association between effective horizontal communication and performance will be strongest under high uncertainty and high equivocality and weakest under low uncertainty and low equivocality.*

Table 1 summarizes the hypotheses concerning the different integration mechanisms for the four cells of the uncertainty–equivocality matrix. When a mechanism appears in bold type, we predict that this is the most appropriate context for it. When it appears in regular type, the mechanism is considered appropriate, but less so than in the previous case. When a mechanism does not appear in a given cell, it is considered least useful in this context according to the theory.

4. Methods

4.1. Research site, sampling and data collection

A cross-sectional survey design was used to test the hypotheses. The data were collected from a single research center of a large corporation. Within this site (called

Centerlab), the sample consisted of 121 different R&D projects from 13 research units (sample sizes are less than 121 in some analyses because of missing data). The research units differed in terms of clients, products and functional specializations based on a variety of fundamental disciplines (i.e., molecular chemistry, robotics, mechanical engineering, nuclear physics, etc.). The projects included technical studies, process development projects and product development projects for both internal and external clients. The choice of a single research site simplifies data collection and controls for a number of possible confounding variables associated with the industry and environment. It does, however, potentially limit the variance in project management procedures including integration mechanisms. Prior to formal data collection, we conducted a number of preliminary interviews to examine the potential diversity of projects, procedures and clients. These appeared to be more than sufficient to render the proposed research meaningful.

The sample of projects was constituted by a snowball sampling method aimed at including all the most important recently completed projects as well as an equal number of relative successes and failures. To be included, projects also had to involve several researchers. At each research unit, the first author began by contacting the Director and administrative assistant. Based on a list of research personnel, all the most senior researchers were then contacted and asked to supply information on one successful and one less successful project during a personal interview. At the end of the interview, the researcher was asked to provide the names of all others who had worked on one or both of these projects. These individuals were then contacted and asked for information on additional projects. Through this process, 116 personal interviews were completed. A questionnaire was left with the respondent after the interview. A further 17 people completed this instrument without a formal interview for a total of 133 questionnaires.

The interview and the questionnaires tapped different variables and all the questionnaires were filled out under similar conditions. Respondents usually completed the questionnaire for two projects (one success and one failure). Most of the researchers excluded from the final sample were younger and less-experienced and were unable to identify two completed projects that fit the inclusion criteria. Ninety percent of those asked to fill out a questionnaire agreed to do so. The high response rate was no doubt assisted by the availability of the first researcher on site for several months. More than one response was obtained for 48% of the project sample (biased towards the larger and more complex projects). In addition, for each project, certain additional information concerning project characteristics was obtained in an interview with the Director of the unit responsible. Table 2 provides some descriptive statistics on the projects in the sample.

4.2. Data collection instruments and measures

The questionnaire administered to participating researchers was the main data collection instrument. This was developed from an analysis of previous literature and where possible, existing scales were adapted to our needs. Before finalization, the questionnaire was pretested with several academics, three researchers from a similar research laboratory and three people from the participating organization.

Table 2
Descriptive statistics on research and development projects in the sample

	All projects		Low uncertainty, low equivocality		Low uncertainty, high equivocality		High uncertainty, low equivocality		High uncertainty, high equivocality	
	Mean (SD)	Median	Mean (SD)	Median	Mean (SD)	Median	Mean (SD)	Median	Mean (SD)	Median
Number of participants	7.0 (9.5)	5	4.0 (2.0)	4	4.0 (1.7)	4	12.4 (19.4)	7	9.6 (4.6)	7
Number of externals	1.2 (2.4)	0	0.5 (1.1)	0	1.0 (2.1)	0	1.5 (3.4)	0	1.9 (2.7)	0.5
Number of respondents	1.8 (1.1)	1	1.4 (0.8)	1	1.5 (0.6)	1	2.0 (1.0)	2	2.3 (1.4)	2
Duration (months)	36 (24)	36	29 (17)	21	46 (30)	36	32 (20)	34	47 (25)	48
Cost (in thousand US\$)	770 (1230)	358	295 (348)	175	517 (585)	300	1205 (2012)	417	1102 (1300)	600
Projects by type	Number (%)		Number (%)		Number (%)		Number (%)		Number (%)	
Technical study	43 (35.5%)		8 (28.6%)		14 (40.0%)		14 (58.3%)		7 (20.6%)	
Process	23 (19.0%)		8 (28.6%)		6 (17.1%)		3 (12.5%)		6 (17.6%)	
Product (int)	28 (23.1%)		7 (25.0%)		11 (31.4%)		2 (8.3%)		8 (23.5%)	
Product (ext)	27 (22.3%)		5 (17.9%)		4 (11.4%)		5 (20.8%)		13 (38.2%)	
TOTAL	121		28		35		24		34	

Four groups of variables are pertinent to the present study. These include the information processing context (project uncertainty and equivocality), the use of integration mechanisms (horizontal structures, formal leadership, planning and process specification, information technology, and informal leadership), horizontal communication effectiveness, and project performance. Each category of variables is examined in turn.

4.2.1. Information processing context

Measures of uncertainty and equivocality were constructed to reflect the dimensions of task variety and analyzability described earlier and used in previous research (see Tushman and Nadler, 1978; Gales et al., 1992; Keller, 1994). These variables were derived from items in the questionnaire combined with responses to questions raised in interviews with senior researchers and unit directors.

The uncertainty dimension reflects the number of information bases and the technical and administrative complexity of the project. It combines three indicators:

- The logarithm of the number of project participants. The number of participants was obtained from the senior researcher for each project by asking him or her to list their names. For 83 projects, this indicator was checked in an interview with the Director of the research unit (the correlation between the two measures was $r = 0.57$).
- Assessments of the administrative complexity and technical complexity, respectively of the project on two 7-point scales obtained from the Director of the research unit.

This indicator of uncertainty (or task variety) is constructed somewhat differently from that proposed by Withey et al. (1983). The items in their scale in fact tend to reflect individual-level task variety rather than task variety across an entire project (Withey et al., 1983: 59). Our measure has more similarity with that of Gales et al. (1992) in the sense that it reflects the number of information bases involved at project-level. This approach applies best to a R&D context where by definition project participants do non-routine tasks and where variety will be reflected more in the multiplicity and diversity of components within the project team than in measures of non-routineness for individual members. (In fact, averaging individual perceptions of personal task-variety may produce little variation between projects). Technical complexity, administrative complexity and the number of research team members provide a better proxy for project-level task variety. Multiple researchers is both a reflection of high complexity and a factor that in itself leads to diversity in bases of information.

The equivocality dimension reflects the newness of the project and its degree of radicality (see also Gales et al., 1992; Gales and Mansour-Cole 1995; Green et al., 1995). It is a composite of five indicators:

- An assessment of whether the project is considered to be “radical” (both objectives and technological methods are not known), “significant” (the product to be developed is unknown but technologies are familiar or vice-versa), or “incremental” (considered as a recombination of well-known methods and techniques towards a well-defined objective). This assessment was obtained from the senior researcher involved in the project.

- Four Likert-type items measuring the novelty of the project for the organization based on Chang and Yong's (1991) measure of technical risk. The items are listed in the Appendix and were included in the main questionnaire to all project participants.

To verify the plausibility of the two distinct dimensions, a factor analysis including all eight items listed above was carried out. Two orthogonal factors with Eigenvalues greater than 1 emerged, explaining 28.8% and 23.7% of the variance, respectively. The last five items all loaded onto the first factor, while the first three all loaded onto the second. Each of the individual indicators was normalized and means were calculated to compose the two scales. Cronbach alphas for uncertainty and equivocality were 0.68 and 0.66, respectively. The scale means (at zero because of normalization) were then used as natural cut-off points to create the four-cell matrix of Table 1. To avoid information loss, the continuous versions of the variables were also used in hypothesis testing.

4.2.2. Integration mechanisms

The multi-item Likert scales used to measure four of the five types of integration mechanisms are given in the Appendix. They were developed for this research, although they were partly inspired by the work of Moenaert and Souder (1990), De Meyer (1991), Préfontaine (1994), and other researchers cited in Section 3. The fifth integration mechanism (informal leadership) was measured using a scale for the use of project champions developed by Zirger (1991). Cronbach alphas for the scales for horizontal structures, formal leadership, planning and process specification and informal leadership were 0.68, 0.68, 0.79, and 0.87, respectively. As can be seen from Table 3, there is a strong correlation between the horizontal structures and the formal planning and process specification variables. We shall comment later on this. All these measures were collected directly from the participants in the research projects using the main questionnaire instrument.

4.2.3. Horizontal communication

The scale for measuring horizontal communication is adapted directly from that proposed by Downs (1988). The complete instrument includes eight dimensions of organizational communication of which five were used in this research. Only the dimension related to horizontal communication is pertinent to this paper. This instrument was chosen because of its relevance to the issue of integration, its simplicity and its recognized psychometric properties (Crino and White, 1981; Greenbaum et al., 1988). The items measuring horizontal communication are listed in the Appendix. Cronbach alpha was 0.85.

4.2.4. Performance

As indicated earlier, all projects were assessed by the senior researcher involved as relative successes or failures. In addition, all project participants were asked to evaluate the project on scales developed by Pinto and Pinto (1990) and Pinto and Slevin (1988) to assess "task outcomes" (seven items). The items used are listed in the Appendix. Cronbach alpha was 0.84 for this scale. Finally, for an independent assessment of performance, the director of each research unit was asked to evaluate all the projects studied from his or her unit on a single-item 7-point scale. The correlation between this

Table 3

Correlations and descriptive statistics for research variables (full sample)
n varies between 115 and 121 because of missing data.

Variables	Mean (SD)	IRA	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Performance	5.17 (1.1)	0.77	(0.84)								
(2) Horiz. structures	2.20 (1.0)	0.59	0.06	(0.68)							
(3) Formal leadersh	5.12 (1.0)	0.62	0.51***	0.07	(0.68)						
(4) Planning	4.16 (1.0)	0.58	0.30***	0.54***	0.19**	(0.79)					
(5) Inf. technologies	3.42 (1.3)	–	0.13	0.28***	–0.05	0.14	–				
(6) Informal leader	3.85 (1.5)	0.56	0.14	0.31***	0.02	0.27***	–0.08	(0.87)			
(7) Horiz. communic	5.02 (1.1)	0.66	0.51***	0.23**	0.39***	0.35***	0.23**	0.12	(0.85)		
(8) Uncertainty	0.00 (0.8)	–	–0.02	0.23**	–0.07	0.19**	0.00	0.19**	0.09	(0.68)	
(9) Equivocality	0.00 (0.7)	–	–0.02	0.05	–0.14	0.00	0.08	0.09	–0.02	0.06	(0.66)

* $p < 0.10$ (two-tailed).

** $p < 0.05$.

*** $p < 0.01$.

item and the main scale was 0.49 indicating moderate levels of agreement between unit directors and researchers. In this study, the researchers' assessment was used as the principle measure because of its stronger foundation in a multi-item scale. Future research should develop measures based on client assessments of performance.

Multiple responses were obtained for 48% of the projects (2 responses for 29%, 3 for 12%, 4 for 5%, 5 for 1.4%, and 6 for 0.7%). This enabled us to evaluate inter-rater agreement (IRA). The measure of IRA is that proposed by James et al. [from James et al. (1984) and James and Demaree (1993)]. Average values range from 0.56 to 0.77 and are presented in Table 3 for all variables for which the assumptions of the method were satisfied. The aggregation of multiple responses then proceeded as follows. The research questionnaire included a question which asked respondents to assess their degree of familiarity with each project on a 7-point scale. This value was used in calculating a weighted average score for all respondents assessing that project. If any respondent's familiarity score was under 4 out of 7, his or her assessment was not included in the aggregate score.

Table 3 provides descriptive data on means and standard deviations of all variables, Cronbach alphas (in diagonal), IRA, and zero order correlations.

4.3. Analysis

Multiple regression was used to test the hypotheses concerning the overall association between the integration variables and performance and to test the mediating effect of horizontal communication (H1a–H6a; H6b). Multiple regression was also used to examine the effects of information processing context on the use of each integration mechanism (H1b–H5b). Moderated regression, correlational analysis and Fisher tests for differences in correlations were used to test hypotheses concerning differences in the degree of association between integration mechanisms and performance for the four cells of the uncertainty–equivocality matrix (H1c–H5c; H6b). As Venkatraman (1989) notes, the use of subgroups and comparisons of correlation coefficients is preferred to moderated regression when the objective is to test the strength rather than the form of moderating effects, as in our hypotheses. We added moderated regression analyses to validate the existence of moderating effects and to provide a test based on all the information available (since subgroup analysis results in information loss).

5. Results

5.1. Integration mechanisms, horizontal communication, and project performance (H1a–H6a; H6b)

The data suggest that several of the integration mechanisms studied contribute to greater project performance. The regression equations presented in Table 4 summarize these results. Model 1 shows the direct effects of the integration mechanisms on performance, Model 2 adds horizontal communication, while Model 3 regresses the integration mechanisms onto horizontal communication as the dependent variable. The

Table 4

Multiple regressions between performance, integration mechanisms and horizontal communication the numbers in the table are standardized regression coefficients.

	Model 1	Model 2	Model 3
Dependent variable	Performance	Performance	Horizontal communication
Independent variables:			
Horizontal structures	−0.23**	−0.24**	0.03
Formal leadership	0.48***	0.37***	0.33***
Planning and process spec.	0.28***	0.20**	0.24**
Information technology	0.18**	0.11	0.21**
Informal leadership	0.13	0.12	0.04
Horizontal communication		0.32***	
<i>F</i> -statistic	12.37***	13.89***	8.59***
<i>n</i>	115	115	115
Adjusted <i>R</i> -squared	0.33	0.40	0.25

* $p < 0.10$

** $p < 0.05$.

*** $p < 0.01$.

results for Models 1 and 2 indicate that formal leadership, planning and process specification and the use of information technologies are significantly associated with performance, supporting hypotheses H2a, H3a and H4a, while the presence of a champion has a positive but insignificant effect, leading us to reject hypothesis H5a. As hypothesized, horizontal communication is also very strongly associated with project performance supporting hypothesis H6a.

Surprisingly, contrary to hypothesis H1a, the coefficient for horizontal structures emerges as significant but negative in Models 1 and 2, suggesting that the use of horizontal structures is detrimental to project performance. This may be an indication of the costs and difficulties of implementing successful horizontal structures noted by other researchers (e.g., Ford and Randolph, 1992; Hitt et al., 1993). However, the discrepancy between this result and the raw correlation coefficient ($r = 0.06$ in Table 3) raises a question as to whether this might also be due to multicollinearity within the data base. There is indeed a relatively high correlation between horizontal structures and the use of planning and process specification in this sample ($r = 0.54$). However, the variance inflation factors remain well within acceptable range for these regressions suggesting that multicollinearity is not a major problem. The overall evidence thus appears to support a somewhat negative effect of horizontal structures on project performance when controlling for the effect of the other independent variables.

The three models also enable us to examine whether horizontal communication plays the role of a mediating variable between the use of integration mechanisms and performance as proposed in hypothesis H6b (see Baron and Kenny, 1986; Venkatraman, 1989). Using the criteria recommended by Baron and Kenny (1986), the results suggest that the effect of the integration mechanisms is partly but not totally mediated by horizontal communication. Model 2 in which horizontal communication is added to the integration mechanisms as an independent variable explains substantially more variance in project performance than Model 1 (an increase of 7%; $p < 0.01$). The standardized

Table 5

Use of integration mechanisms by uncertainty and equivocality

		Low uncertainty (low task variety)		High uncertainty (high task variety)		
		Mean	SD		Mean	SD
High equivocality (low task analyzability)	Performance	5.4	1.0	Performance	5.1	1.2
	Horizontal structures	1.8	1.0	Horizontal structures	2.4	1.1
	Formal leadership	5.4	1.1	Formal leadership	5.1	1.4
	Planning and process spec.	3.9	1.2	Planning and process spec.	4.4	1.0
	Information technology	3.7	1.3	Information technology	3.5	1.3
	Informal leadership	3.7	1.6	Informal leadership	4.3	1.4
	Horizontal communication (<i>n</i> = 35)	5.0	1.3	Horizontal communication (<i>n</i> = 34)	5.2	1.0
Low equivocality (high task analyzability)	Performance	5.1	0.9	Performance	5.1	0.9
	Horizontal structures	2.1	1.0	Horizontal structures	2.3	0.9
	Formal leadership	5.4	1.1	Formal leadership	5.3	1.1
	Planning and process spec.	4.2	1.2	Planning and process spec.	4.5	0.8
	Information technology	3.2	1.4	Information technology	3.3	0.9
	Informal leadership	3.5	1.2	Informal leadership	4.2	1.5
	Horizontal communication (<i>n</i> = 28)	4.9	0.9	Horizontal communication (<i>n</i> = 24)	5.0	1.0

beta coefficients for the individual mechanisms are reduced somewhat, but except for information technologies, they remain significant. Also, the regression equation relating integration mechanisms to horizontal communication is significant but explains only 25% of the variance in horizontal communication. We conclude that the integration mechanisms of planning and process specification and formal leadership have both a direct and an indirect effect on project performance, but that the effect of information technologies is largely indirect, mediated by horizontal communication. In addition, horizontal communication has a unique effect on project performance, independently of the integration mechanisms used.

Table 6

Regression analysis of integration mechanisms on uncertainty and equivocality

Dependent variables	Horizontal communication	Horizontal structures	Formal leadership	Planning and process spec	Information technology	Informal leadership
Context variables:						
Main effects						
Uncertainty	0.10	0.23**	−0.06	0.17*	−0.01	0.19**
Equivocality	−0.01	0.02	−0.10	−0.07	0.07	0.05
Interaction						
Uncertainty × Equivocality	0.03	0.21**	−0.04	0.15	−0.04	0.07
Adjusted <i>R</i> -squared	0.01	0.10	0.02	0.05	0.00	0.05

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$. NS $p > 0.10$; numbers in the table are standardized betas.

Table 7
Multiple regressions between performance and integration mechanisms with moderating effects of context
NS, $p > 0.10$. The numbers in the table are standardized regression coefficients.

	Model 2 (see Table 6)	Model 4 (Model 2 with context controls)	Model 5 (Model 4 + uncertainty interactions)	Model 6 (Model 4 + equivocality interactions)	Model 7 (Model 4 + all two-way interactions)	Model 8 (Model 4 + two- and three-way interactions)
Independent variables						
Horizontal structures	−0.24**	−0.23**	−0.23**	−0.16	−0.17*	−0.24**
Formal leadership	0.37***	0.36***	0.34***	0.37***	0.36***	0.35***
Planning and process spec.	0.20**	0.20**	0.20**	0.18***	0.16*	0.24**
Information technology	0.11	0.10	0.10	0.08	0.07	0.12
Informal leadership	0.12	0.13	0.14	0.07	0.11	0.10
Horizontal communication	0.32***	0.33***	0.30***	0.30***	0.29***	0.24**
Information-processing context						
Uncertainty		−0.06	−0.49	−0.07	−0.92*	−0.90
Equivocality		0.04	−0.04	−0.75	−0.77	−0.21
Uncertainty × equivocality						0.41
Two-way interactions						
Uncertainty ×						
Horizontal structures			0.23		0.05	0.08
Formal leadership			0.71*		0.81*	1.13**
Planning and process spec.			−0.26		−0.26	−0.28
Information technology			−0.17		−0.19	0.18
Informal leadership			−0.25		−0.16	−0.19
Horizontal communication			0.18		0.61	−0.03

Equivocality ×						
Horizontal structures				0.01	−0.07	0.07
Formal leadership				−0.91**	−1.18**	−1.41**
Planning and process spec.				−0.18	0.09	−0.42
Information technology				0.37	0.36	0.24
Informal leadership				0.06	0.08	0.12
Horizontal communication				1.38**	1.50***	1.67***
Three-way interactions						
Uncertainty × equivocality ×						
Horizontal structures						0.22
Formal leadership						−0.77
Planning and process spec.						−1.24**
Information technology						−0.18
Informal leadership						−0.55*
Horizontal communication						2.18**
R-squared	0.433	0.438	0.467	0.501	0.539	0.602
Adjusted R-squared	0.402	0.395	0.393	0.432	0.441	0.481
Δ R-squared		0.005	0.029	0.063	0.101	0.044
			(vs. Model 4)	(vs. Model 4)	(vs. Model 4)	(vs. Model 7 + inc × equ.)
F − Δ R-squared		0.476 (NS)	0.916 (NS)	2.125*	1.734*	1.621 (NS)

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.

5.2. Uncertainty, equivocality and the use of integration mechanisms (H1b–H5b)

The data suggest that the company does not adjust its integration mechanisms to information processing needs in quite the way predicted by the Daft and Lengel (1986) model. Table 5 shows the means and standard deviations of the integration variables for each of the four cells of the uncertainty–equivocality matrix. To test hypotheses H1b–H5b, which reflect the “selection” approach to the Daft and Lengel theory, we regressed each of the integration mechanisms on uncertainty, equivocality and their interaction. The results are shown in Table 6.

We expected that horizontal structures would be used more frequently for projects with high equivocality (H1b). However, the analysis in Tables 5 and 6 suggests that structural mechanisms are more likely to be introduced to cope with high uncertainty. Contrary to our hypothesis, projects that are low in uncertainty but high in equivocality are least likely to involve horizontal structures (Table 5). This is reflected in the significant interaction term in Table 6.

We predicted that both uncertainty and equivocality would affect the use of planning and process specification (H3b). The analysis shows a small positive effect for uncertainty, suggesting that planning and process specification is seen by Centerlab principally as a means to cope with uncertainty rather than equivocality.

The use of informal leadership (project champions) was expected to be lower for projects of high uncertainty given the difficulties of exerting influence with little formal authority in these circumstances. The data suggest the opposite: project championship is in fact more likely for more uncertain projects.

Finally, there are no significant differences in the use of formal leadership or information technology among the different cells. It appears that Centerlab pays relatively little attention to the project context in determining management approaches despite considerable variance in project, size and complexity (see Table 2).

It is also worth noting that uncertainty and equivocality do not appear to have a significant negative effect on performance overall (see Tables 3 and 5). Comments made by managers during interviews tend to suggest that while more complex and novel projects are indeed more risky, the increased attention accorded to them compensates partly for this effect. Smaller and more incremental projects may be easier to handle, but they can also suffer from relative neglect.

5.3. Uncertainty, equivocality and the differential effect of integration mechanisms on performance (H1c–H6c)

The data provide some support for the proposition that the effect of integration mechanisms on performance will vary depending on uncertainty and equivocality (hypotheses H1c–H6c) although the pattern of relationships is not exactly as predicted. Table 7 presents moderated regression equations relating performance to integration mechanisms, uncertainty and equivocality with two- and three-way interactions. Model 4 adds the direct effects of uncertainty and equivocality to Model 3 (reproduced from Table 4). Model 7 shows that the inclusion of first-order moderating effects for the context explains an additional 10% of variance ($p < 0.10$).

A closer look at the moderating effects of uncertainty and equivocality taken separately (Models 5 and 6) indicates that the addition of equivocality interactions alone significantly improves the explanation of performance while the inclusion of uncertainty interactions alone does not. Because the effects of uncertainty and equivocality are not necessarily independent of one another, we also added three-way interaction effects in Model 8. Despite several significant coefficients, this model falls just short of adding significant explanatory power.

To better understand the nature of the contingency effects for each integration mechanism, correlations between integration variables and performance for the four cells of the uncertainty–equivocality matrix are shown in Table 8. Table 9 presents significant differences among these. We now combine these results with those of Table 7 to examine each of our hypotheses in turn. Recall that hypotheses H1c–H6c are based on the “interaction” interpretation of Daft and Lengel’s theory (Van de Ven and Drazin, 1985) in which it is assumed that a better fit between integration mechanisms and information processing context will result in higher performance.

Table 8

Correlations between integration variables and performance by uncertainty and equivocality

Letters in bold indicate that this variable is statistically more highly correlated with performance in that cell than in others (Fisher test for differences in correlations with $p < 0.10$, one-tailed test).

	Low uncertainty (low task variety)		High uncertainty (high task variety)	
	Corr	Sig	Corr	Sig
High equivocality (low task analyzability)	(1) uE		(2) UE	
Horizontal structures	0.13	–	Horizontal structures	0.20 –
Formal leadership	0.20	–	Formal leadership	0.65 ***
Planning and proc spec.	0.50	***	Planning and proc spec.	0.30 *
Information technology	0.14	–	Information technology	0.31 *
Informal leadership	0.41	**	Informal leadership	0.07 –
Horizontal communication	0.64	***	Horizontal communication	0.71 ***
($n = 35$)			($n = 34$)	
Low equivocality (high task analyzability)	(3) ue		(4) Ue	
Horizontal structures	–0.12	–	Horizontal structures	0.00 –
Formal leadership	0.54	***	Formal leadership	0.64 ***
Planning and process spec.	0.13	–	Planning and process spec.	0.40 *
Information technology	–0.15	–	Information technology	0.04 –
Informal leadership	–0.30	–	Informal leadership	0.24 –
Horizontal communication	0.18	–	Horizontal communication	0.31
($n = 28$)			($n = 24$)	

* $p < 0.10$ (two-tailed significance).

** $p < 0.05$.

*** $p < 0.01$.

Table 9
Significant differences in correlations between integration variables and performance for different levels of uncertainty and equivocality

Variable	Significant differences in correlations with performance between cells ($p < 0.10$ — one-tailed test). (Cell key in Table 8)
Horizontal structures	none*
Formal leadership	uE < UE; uE < ue; uE < Ue
Planning and process specification	ue < uE
Information technology	ue < UE
Informal leadership	ue < uE; ue < Ue; ue < UE
Horizontal communication	ue < uE; ue < UE

* But not that the test for differences between cells ue and UE gives a p of 0.102, just under the threshold level.

For horizontal structures, the results of the moderated regression analysis of Table 7 suggest no significant effects of project context. While the pattern of bivariate correlations in Table 8 is not inconsistent with the prediction that the use of horizontal structures will be more positively related to project performance under high equivocality than under low equivocality (H1c), these relationships are not significant nor detectable in the multivariate analysis of Table 7. The benefits expected from horizontal structures in achieving integration for projects of high equivocality do not emerge clearly in this sample.

For formal leadership, we predicted that the association between the use of formal project leadership and performance would be strongest for projects of high equivocality and high uncertainty and weakest for projects of low equivocality and low uncertainty (H2c). The moderated regressions do indeed indicate that the impact of formal leadership varies depending on the context. However, the sign of the interaction term is contrary to our predictions, suggesting that equivocality dampens rather than enhances the relationship between formal leadership and performance. Table 8 throws further light on these results, showing high correlations between formal leadership and performance in all contexts except low uncertainty and high equivocality. We return

For planning and process specification, we predicted that planning would be most effective in enhancing performance in contexts of high uncertainty and high equivocality and least effective in contexts of low uncertainty and low equivocality (H3c). The corresponding two-way interactions in Table 7 are not significant, providing no support for this hypothesis. The correlation coefficients in Table 8 do however tend to suggest that the association between planning and performance is lowest under low uncertainty and low equivocality as we predicted.

Table 8 suggests that the relationship between the use of information technology and performance is significantly positive only under high equivocality and high uncertainty. However, none of the interaction effects in the moderated regressions are significant offering little support for our hypothesis that the benefits of information technology are greater under high uncertainty (H4c).

We predicted that informal leadership (presence of a project champion) would be most strongly associated with performance under conditions of high equivocality and low uncertainty (H5c). This is supported by the data in Table 8 and is also reflected in the three-way interaction coefficients of Table 7 (Model 8).

Finally, the pattern of relationships between horizontal communication and performance supports hypothesis (H6b). In particular, the moderated regressions show that equivocality increases the importance of horizontal communication for performance (significantly positive interaction term). Tables 8 and 9 also indicates that effective horizontal communication is most strongly related to project performance under high uncertainty and high equivocality and least related to performance under low uncertainty and low equivocality.

In fact, over and above the tests of specific hypotheses, one additional observation emerges from the correlations and associated tests in Tables 8 and 9. This is that with the exception of formal leadership, all integration mechanisms tend to be less positively related to performance for projects of low uncertainty and low equivocality than in certain other cells.

6. Discussion and conclusions

This research aimed to contribute to a better understanding of the role of integration mechanisms in R&D project success. The information processing perspective (Galbraith, 1973; Tushman, 1979; Daft and Lengel, 1986; Gales et al., 1992) was adopted as a coherent theoretical framework for understanding the contingency relationships affecting the appropriateness of different mechanisms. Table 10 synthesizes the results obtained for the full set of hypotheses. In this section, we highlight the main conclusions, discuss certain surprising results, note the limitations of this research, and examine the implications of our study for research and practice.

6.1. *Integration mechanisms and project performance*

First, we confirmed that several integration mechanisms discussed in the literature can contribute to project success. In particular, there is evidence that empowered project leadership is critical. This supports empirical observations by Ancona and Caldwell (1990) and Clark and Fujimoto (1991). However, our data suggest that the importance of strong formal leadership is not restricted to projects of high uncertainty or high equivocality as we predicted based on information processing theory. This means that “integration” of many diverse specialists (Daft and Lengel, 1986) is probably not the only important role of these leaders. We may speculate, for example, that in the case of smaller and simpler projects, powerful leaders may contribute to success through their capacity to attract resources to projects that might otherwise be neglected. The only context in which formal leadership was not significantly related to performance was the low uncertainty, high equivocality condition. This result is more difficult to explain.

Table 10
Synthesis of results for all hypotheses

Integrating mechanism	Direct effects on performance	Contingency effect of context (selection model)*	Contingency effect of context (interaction model)
Horizontal structures	H1a: not supported. Negative relationship	H1b: not supported. Use of horizontal structures increases with uncertainty, but not with equivocality	H1c: not supported. Context does not moderate effects of horizontal structures on performance
Formal leadership	H2a: strongly supported. Positive relationship	H2b: not supported. Use of formal leadership does not vary by context	H2c: partially supported. Context moderates effects of leadership on performance. Surprisingly low impact for low uncertainty, high equivocality
Planning and process specification	H3a: strongly supported. Positive relationship	H3b: partially supported. Use of planning increases with uncertainty	H3c: weakly supported. Effect of planning on performance is least for low uncertainty, low equivocality
Information technology	H4a: supported. Positive relationship	H4b: not supported. Use of IT does not vary by context	H4c: little support. No interaction effects, but effect of IT on performance is strongest for high uncertainty, high equivocality

Informal leadership	H5a: not supported. Non-significant relationship	H5b: not supported. Use of champions is not related to equivocality but increases with uncertainty	H5c: partially supported. Effect of champion on performance is least for low uncertainty, low equivocality
Horizontal communication	H6a: strongly supported. Positive relationship	H6b (mediator hypothesis): supported. Horizontal communication partly mediates relationships between integration mechanisms and performance.	H6c: strongly supported. Context (especially equivocality) moderates performance effects of horizontal communication

* N.B. Hypothesis H6b does not deal with contingency effects but with the mediating effect of horizontal communication.

The important effect of informal leadership (project championship) under this condition (see Table 8) but not in others suggests that there may be some substitution between ideal forms of leadership here. In other words, informal spontaneous leadership may be more appropriate for small projects when there is ambiguity concerning project technology because it allows the most qualified member of the team to dominate decisions. Attempts by formal leaders designated by management to take control may be counter-productive in such circumstances. Leaders may also be reluctant to use their influence in such projects because the risks far outweigh the potential returns.³

We also found that planning and process specification is positively related to performance under all conditions except when uncertainty and equivocality are low. This confirms Daft and Lengel's (1986) conception of these processes as relevant to both uncertainty and equivocality reduction. Our data also provide some limited evidence of the usefulness of information technology as an integration device. Its effect appears to be almost entirely due to improved horizontal communication effectiveness. As shown, in Table 8, the effect is also apparently greatest for projects of high uncertainty and equivocality. As proposed by more recent empirical research (Fulk and Boyd, 1991; Hinds and Kiesler, 1995), all this tends to suggest that these technologies may be able to support richer forms of information exchange than previously believed.

Despite a great deal of emphasis in the literature on the importance of horizontal structures for enhancing integration (Brown and Eisenhardt, 1995), we found almost no evidence for their performance benefits in our sample, and some indications that they may hurt performance. As noted by Ford and Randolph (1992) and Hitt et al. (1993), these devices are difficult to use well. While they may have the potential to enhance coordination, they can also be costly. Ambiguous reporting relationships created by such structures may lead to conflict that can cause breakdowns in communication.⁴ We predicted that projects of high uncertainty and equivocality would provide the most propitious context for horizontal structures. However, the data do not provide a clear demonstration that they are worthwhile even here. One explanation for this may lie in the rather low level of experimentation with horizontal structures by the firm studied (see Table 5).

The data do, however, confirm the importance of effective horizontal communication for project performance, especially for more innovative projects (e.g., those of high equivocality) (see also Tushman, 1979; Allen, 1986; Gales et al., 1992; Brown and Eisenhardt, 1995; Moenaert et al., 1995). As expected, horizontal communication is a partial mediator of the relationship between integration mechanisms and performance. However, the predictive power of integration mechanisms on horizontal communication

³ Thanks to an anonymous reviewer for suggesting this point.

⁴ Following a suggestion by one of the reviewers, we introduced quadratic terms for both the horizontal structures and the planning and process specification variables into the regression equations of Models 1 and 2 to determine whether there might be a curvilinear relationship between these mechanisms and performance (implying the existence of some optimal level at which the costs and benefits would be perfectly balanced out). The coefficients were not significant ($p > 0.10$) in both cases. There is clearly a need for further research on these relationships.

is more limited, and both sets of variables appear to have independent direct effects on performance. In practical terms, this means two things:

First and most obviously, effective horizontal communication appears at least partly to be a spontaneous activity which may be difficult for a firm to control or predict simply by applying formal structural mechanisms. As Dougherty and Hardy (1996) have pointed out, many firms have had difficulty reproducing successful product innovation efforts. This may be partly because their previous success is due more to the chemistry between specific team members than to formal structures and procedures. To be able to sustain innovation, firms must develop organizational cultures that favor the kind of collaborative relationships associated with success, with horizontal structures serving a facilitating but not dominant role (Dougherty, 1992; Hitt et al., 1993).

Secondly, the results imply that the chosen integration mechanisms affect performance independently of their effect on horizontal communication. One way in which mechanisms such as planning and information systems may improve performance is, in fact, by helping to avoid the necessity for extensive communication. As we noted in our presentation of the theory, the break-down of projects into simple component parts enable each participant to execute their portion of the project fairly independently (Moenaert and Souder, 1990; Von Hippel, 1990; Wheelwright and Clark, 1992). The milestones, explicit goals, and evaluatory procedures associated with planning and process specification may also have a motivational effect in stimulating efficient project activities (Locke and Latham, 1990). In addition, as noted earlier, both formal and informal leadership may contribute by facilitating resource acquisition and by enhancing motivation as well as through their ability to improve integration and horizontal communications.

6.2. Degree of support for the information processing perspective

Over and above specific observations concerning the performance effects of various integration mechanisms, how much overall support did we find for the information processing perspective? Hypotheses H1b–H5b reflect the “selection” approach to the theory, which suggests that integration mechanisms will be used more extensively when they are “appropriate” to the context. We found rather limited support for these hypotheses. Centerlab did not tend to modulate its use of integration mechanisms as a function of project equivocality. Moreover, its adjustments for project uncertainty (for planning, horizontal structures and informal leadership) were not always in the predicted directions. These results tend to suggest that R&D laboratories may have some difficulty in adjusting their project management approaches to a wide range of different situations. Internal rules and traditions may tend to lead to some uniformity. When adjustments are made, they may be more readily justified by tangible factors such as the complexity of the project and the number of different specialties involved (factors related to project uncertainty) rather than by less tangible considerations such as the novelty of the product or process being developed (project equivocality). One hypothesis worthy of future research is that while task uncertainty is treated as an organizational problem, equivocality is more of an individual problem which will be handled by individual-level rather than organizational-level strategies.

The alternative approach to contingency theory is the “interaction” perspective (Van de Ven and Drazin, 1985) which posits that the closer the fit between context and behavior, the higher the performance. This approach was reflected in hypotheses H1c–H6c. We find a modest level of support for this form of the theory. In fact, there is evidence that equivocality moderates the performance effects of integration devices to a greater extent than uncertainty. Formal leadership and horizontal communications are the two mechanisms that show the strongest effects here, despite the fact that their use did not emerge as significantly affected by context in the “selection” approach analysis.

This suggests the possibility of a degree of complementarity between the two approaches to contingency theory. When the firm adjusts its approach to the context, performance effects may be undetectable because unpropitious cases are eliminated beforehand. When it does not do so but should, significant performance effects are more likely to emerge. This argument would lead one to expect that the “selection” approach will be more predictive for variables largely under the control of management while the “interaction” approach will be more predictive for less controllable variables. In our case, the variables horizontal structures (more controllable), planning and process specification (more controllable) and horizontal communication effectiveness (less controllable) fit the pattern. However, the pattern of relationships for informal leadership (less controllable) and formal leadership (more controllable) appear at first to contradict this supposition. Clearly, this could be an interesting avenue for future research.

Another observation is that integration mechanisms are in general least useful for less uncertain and equivocal projects. This means that the factors affecting the performance of less complex and more incremental projects lie elsewhere. In fact, paradoxically, we can explain more of the variance in the performance of complex, large and radical innovations than of smaller and more incremental ones. Again, this suggests a need for more research.

6.3. Strengths and limitations of the research

Research on technological innovation has often been criticized for its largely atheoretical nature (e.g., Brown and Eisenhardt, 1995). The information processing perspective provides one approach to filling this theoretical gap (Gales et al., 1992). A number of recent studies have adopted this perspective for the study of R&D project success. For example, Gales et al. (1992) and Keller (1994) focused on the effects of task variability and analyzability on the amounts of information processed while Gales and Mansour-Cole (1995) examined the relationship between client involvement, project uncertainty and performance. In this study, we attempted to take a further step forward by examining the role of integration mechanisms in relation to uncertainty and equivocality and the mediating effect of horizontal communication (Daft and Lengel, 1986).

Like any other study, ours is not without limitations. Our focus on a single research laboratory allowed the control of certain contextual variables and enabled us to gather data on a fairly large sample of projects through personal interviews and questionnaires from several sources. The use of multiple respondents for each project is relatively unusual in this type of research and enhances the quality of the data base. However, a

single site does limit the generalizability of the research and may have restricted the range of certain variables (e.g., horizontal structures) limiting our capacity to detect significant relationships. Future work should also use measures of performance based on independent client assessments. Our use of researcher assessments of performance may raise a concern about common method bias. To mitigate this, we made an effort to measure this variable using factually formulated questions (see Appendix). Another source of bias could arise from the pooling of data which included more than one project rating from each respondent. This concern is somewhat mitigated by the process of aggregation which took responses from people involved in evaluating different project pairs. Finally, as a cross-sectional field study, this research provides evidence of association, but not causality.

6.4. Implications and directions for research and practice

This study suggests several directions for future work, several of which have been mentioned above. First, there is a need to verify these results in other contexts using similar and different methodologies. For example, client assessments of performance would be useful, while longitudinal methods would help establish causal linkages. In addition, there is a need to test richer explanatory models that incorporate networks of variables. For example, we have seen that horizontal structures may well have counter-vailing effects: better communication on the one hand, but higher cost on the other, producing an ambiguous effect on overall performance (Ford and Randolph, 1992; Hitt et al., 1993). These relationships could be modeled and tested explicitly in different types of contexts using intervening variables and structural equations modeling.

Curvilinear models relating particular integration mechanisms (e.g., planning) to information processing context are also theoretically defensible (e.g., Boynton et al., 1993; Adler, 1995) and should be tested further. Certain integration mechanisms studied also warrant finer definition. For example, formal and informal leaders may have a variety of different styles that could be more or less appropriate in different contexts. Finally, the role of uncertainty and equivocality as significant moderating variables also underlines the need for research on innovation to take these conditions into account when selecting samples for study. The pooling of diverse sets of projects into a single sample potentially contributes to instability in results (Downs and Mohr, 1976).

Our research also offers lessons for managers. First, we note that smaller and simpler projects performed no better than larger and more complex ones despite the head-start they had in principle. Managers should beware of neglecting simpler projects that nevertheless make up a large proportion of the R&D budget. Our data suggest that formal leadership appears to be as important in this context as in others, though probably for different reasons. Here its effects are likely to be related to the capacity to mobilize resources and attention to get the job done. For more uncertain and equivocal projects, the integrative role will be more significant.

Secondly, we note a tendency to adjust project management practices to uncertainty but not to equivocality. This is perhaps partly because uncertainty is more obvious and more easily detectable. Managers need to ensure that the special needs of highly innovative projects are recognized. Another reason for this may be that equivocality is

simply less manageable by formal mechanisms alone. Our results suggest that while formal planning is not without merit in these circumstances, success may be more dependent on individual behavior and the resulting spontaneous chemistry that develops between project participants (see also Dougherty and Hardy, 1996). To handle this, ways need to be found to stimulate effective horizontal communications beyond what can be achieved simply with formal structures, procedures and technologies. This might involve the selection of people to take part in projects taking into account not only their technical prowess, but also their capacity to collaborate, their compatibility with other project participants and their informal leadership potential. This is a task that requires subtlety and foresight, or possibly some form of mutual self-selection. In conclusion, we suspect that softer and more unpredictable management skills need to be combined with formal integration mechanisms to optimize the innovative capability of R&D project groups.

Appendix A. Scales for variables used in the research

Equivocality:

(From senior project researcher) Classification of project as “incremental”, “significant” or “radical”.

“To what extent do you agree with the following statements?”

(R) The product benefitted from its closeness to the company’s existing products

The product was to be the first of its kind in the firm

The product has little similarity to our current products

No one was completely certain about the design/process approach

Uncertainty:

(From unit director) Administrative complexity

(From unit director) Technical complexity

(From senior project researcher) Logarithm of number of project participants

Performance: “To what extent do you agree with the following statements?”

The product/process met quality standards

The technical outcome of the project met expectations

The project was on schedule

All the tasks that we set out to do were accomplished

The project was on or near budget ($\pm 10\%$)

The project’s goals were achieved

The product’s profitability exceeded the average level for this type of investment (however profitability is measured)

Horizontal structures: *“To what extent were the following decision-making mechanisms used?”*

Steering committee

Meetings with department heads

“To what extent do the following methods correspond to the way the projects were organized and directed?”

Use of a cross-functional task force

Use of a matrix organization (double responsibility, usually functional and product)

Temporary assignment of personnel to another department

Formal leadership: *“To what extent do you agree with the following statements?”*

The project manager (or the person officially responsible) had the authority for the project

This person had the necessary skills

This person chose the personnel for the project

Planning and process specification: *“To what extent were the following decision-making mechanisms used?”*

Milestone reports

Formal go/no go gate

Client's representative sign-off on design review

“To what extent were the following methods used to ensure control or communication?”

Financial planning (including projections of future costs and revenues product might produce)

Formal program to upgrade the product development process

Time planning of activities (network scheduling)

Work breakdown structure (activities' planning)

Measure of product development success

Information technology: *“To what extent were the following technologies used during the project?”*

Electronic mail (interpersonal communication by computer)

Voice mail (telephone mailbox)

Facsimile (fax)

Electronic document management

Access to external databases

Computer-aided design (CAD)

Computer-aided engineering (CAE)

Informal leadership (champion): “*To what extent do you agree with the following statements?*”

There was a “product champion” who actively promoted the project within the firm

The “product champion” was high up in the organization

The “product champion” had great influence in matters concerning the project

Horizontal communication: “*To what extent were you satisfied with the following points?*”

During the project, the informal communication network was active in our organization

Horizontal communications with other employees were accurate and free-flowing

Communication practices were adaptable to emergencies

My work group was compatible

Information communications were active and accurate

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